In [1]:
```
# Mengimport syntaks / library yang dibutuhkan
```

In [2]:
```
import os
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold, cross_valida
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, classification_report, ConfusionMatrixDisplay
)

RANDOM_STATE = 42
sns.set_theme(style="whitegrid")
pd.set_option("display.max_columns", None)
print("Library berhasil diimpor.")
```

```
Library berhasil diimpor.
```

In [3]:
```
#Membaca file original.csv dan menampilkan 5 data teratas
```

In [4]:
```
file_path = "original.csv"

if not os.path.exists(file_path):
    try:
        from google.colab import files
        uploaded = files.upload()
        file_path = next(iter(uploaded))
    except Exception:
        raise FileNotFoundError(
            "File original.csv tidak ditemukan. Letakkan original.csv satu folder "
            "dengan notebook, lalu jalankan ulang sel ini."
        )

df_raw = pd.read_csv(file_path)
print("Dataset berhasil dibaca.")
print("Dimensi awal:", df_raw.shape)
display(df_raw.head())
```

```
Dataset berhasil dibaca.
Dimensi awal: (87, 22)
```

	Gender	Age	POLAR4 Quintile	POLAR3 Quintile	Adult HE 2001 Quintile	Adult HE 2011 Quintile	TUNDRA MSOA Quintile	TUNDRA LSOA Quintile	Gaps GCSE Quintile	Gap GCS Ethnici Quinti
0	0.0	38.0	5.0	5.0	4.0	5.0	4.0	4.0	5.0	5
1	0.0	22.0	4.0	2.0	1.0	3.0	4.0	4.0	4.0	5
2	0.0	21.0	4.0	5.0	4.0	5.0	4.0	4.0	5.0	5
3	0.0	25.0	5.0	4.0	3.0	4.0	3.0	3.0	5.0	5
4	0.0	20.0	5.0	4.0	3.0	4.0	3.0	3.0	4.0	4

In [5]:
```
# Menampilkan informasi data dari file original.csv seperti: tipe data, jumlah miss
```

In [6]:
```
print("Informasi tipe data:")
df_raw.info()

print("\\nJumlah nilai hilang per kolom:")
display(df_raw.isna().sum().to_frame("Jumlah Missing Value"))

print("\\nStatistik deskriptif:")
display(df_raw.describe().T)
```

```
Informasi tipe data:
<class 'pandas.DataFrame'>
RangeIndex: 87 entries, 0 to 86
Data columns (total 22 columns):
 #   Column                             Non-Null Count  Dtype  
---  ------                             --------------  -----  
 0   Gender                             80 non-null     float64
 1   Age                                80 non-null     float64
 2   POLAR4 Quintile                    80 non-null     float64
 3   POLAR3 Quintile                    80 non-null     float64
 4   Adult HE 2001 Quintile             80 non-null     float64
 5   Adult HE 2011 Quintile             80 non-null     float64
 6   TUNDRA MSOA Quintile               80 non-null     float64
 7   TUNDRA LSOA Quintile               80 non-null     float64
 8   Gaps GCSE Quintile                 80 non-null     float64
 9   Gaps GCSE Ethnicity Quintile       80 non-null     float64
 10  Uni Connect target ward            80 non-null     float64
 11  attending from home?               80 non-null     float64
 12  distance to university (km)        80 non-null     float64
 13  Count of Module Area Logins        80 non-null     float64
 14  Total Hours in Module Area         80 non-null     float64
 15  % of Average Hours in Module Area  80 non-null     float64
 16  # of presence                      80 non-null     float64
 17  # of Absence                       78 non-null     float64
 18  Percent Attended                   80 non-null     float64
 19  label (fail=1, pass=0)             80 non-null     float64
 20  Unnamed: 20                        0 non-null      float64
 21  Unnamed: 21                        0 non-null      float64
dtypes: float64(22)
memory usage: 15.1 KB
\nJumlah nilai hilang per kolom:
```

	Jumlah Missing Value
Gender	7
Age	7
POLAR4 Quintile	7
POLAR3 Quintile	7
Adult HE 2001 Quintile	7
Adult HE 2011 Quintile	7
TUNDRA MSOA Quintile	7
TUNDRA LSOA Quintile	7
Gaps GCSE Quintile	7
Gaps GCSE Ethnicity Quintile	7
Uni Connect target ward	7
attending from home?	7
distance to university (km)	7
Count of Module Area Logins	7
Total Hours in Module Area	7
% of Average Hours in Module Area	7
# of presence	7
# of Absence	9
Percent Attended	7
label (fail=1, pass=0)	7
Unnamed: 20	87
Unnamed: 21	87

```
\nStatistik deskriptif:
```

	count	mean	std	min	25%	50%	75%	
Gender	80.0	0.125000	0.332805	0.000000	0.000000	0.000000	0.000000	1.000
Age	80.0	22.212500	4.451408	19.000000	20.000000	21.000000	22.000000	44.000
POLAR4 Quintile	80.0	3.575000	1.438837	1.000000	2.000000	4.000000	5.000000	5.000
POLAR3 Quintile	80.0	3.287500	1.433715	1.000000	2.000000	4.000000	4.250000	5.000
Adult HE 2001 Quintile	80.0	2.712500	1.203305	1.000000	2.000000	3.000000	4.000000	5.000
Adult HE 2011 Quintile	80.0	3.275000	1.395971	0.000000	2.000000	4.000000	4.000000	5.000
TUNDRA MSOA Quintile	80.0	2.850000	1.293606	0.000000	2.000000	3.000000	4.000000	5.000
TUNDRA LSOA Quintile	80.0	2.850000	1.293606	0.000000	2.000000	3.000000	4.000000	5.000
Gaps GCSE Quintile	80.0	3.650000	1.543463	0.000000	3.000000	4.000000	5.000000	5.000
Gaps GCSE Ethnicity Quintile	80.0	4.112500	1.449520	0.000000	4.000000	5.000000	5.000000	5.000
Uni Connect target ward	80.0	0.100000	0.301893	0.000000	0.000000	0.000000	0.000000	1.000
attending from home?	80.0	0.662500	0.475840	0.000000	0.000000	1.000000	1.000000	1.000
distance to university (km)	80.0	29.241366	49.616500	0.000000	3.259523	15.711609	32.451973	317.327
Count of Module Area Logins	80.0	50.175000	38.903898	1.000000	30.750000	44.500000	62.000000	335.000
Total Hours in	80.0	18.626000	20.910481	0.030000	8.550000	14.805000	22.237500	175.440

	count	mean	std	min	25%	50%	75%	
Module Area								
% of Average Hours in Module Area	80.0	1.073753	0.846172	0.002513	0.581920	0.862037	1.286354	5.503
# of presence	80.0	8.700000	3.016411	0.000000	7.000000	9.000000	11.000000	13.000
# of Absence	78.0	2.935897	2.870991	0.000000	0.250000	2.000000	4.750000	12.000
Percent Attended	80.0	74.375042	24.750711	0.000000	58.333333	83.333333	100.000000	100.000
label (fail=1, pass=0)	80.0	0.150000	0.359324	0.000000	0.000000	0.000000	0.000000	1.000
Unnamed: 20	0.0	NaN	NaN	NaN	NaN	NaN	NaN	
Unnamed: 21	0.0	NaN	NaN	NaN	NaN	NaN	NaN	

In [7]:
```
#Merapikan nama kolom dari spasi yang berlebihan
```

In [8]:
```
df = df_raw.copy()
df.columns = df.columns.astype(str).str.strip()
```

In [9]:
```
#Merapikan baris/kolom dan validasi target
```

In [10]:
```
df = df.dropna(axis=0, how="all")
df = df.dropna(axis=1, how="all")

target_col = "label (fail=1, pass=0)"
if target_col not in df.columns:
    raise KeyError(f"Kolom target '{target_col}' tidak ditemukan. Kolom tersedia: {
```

In [11]:
```
#Membersihkan baris kosong pada target, menghitung jumlah persentase kelas
```

In [12]:
```
df = df.dropna(subset=[target_col]).copy()
df[target_col] = df[target_col].astype(int)

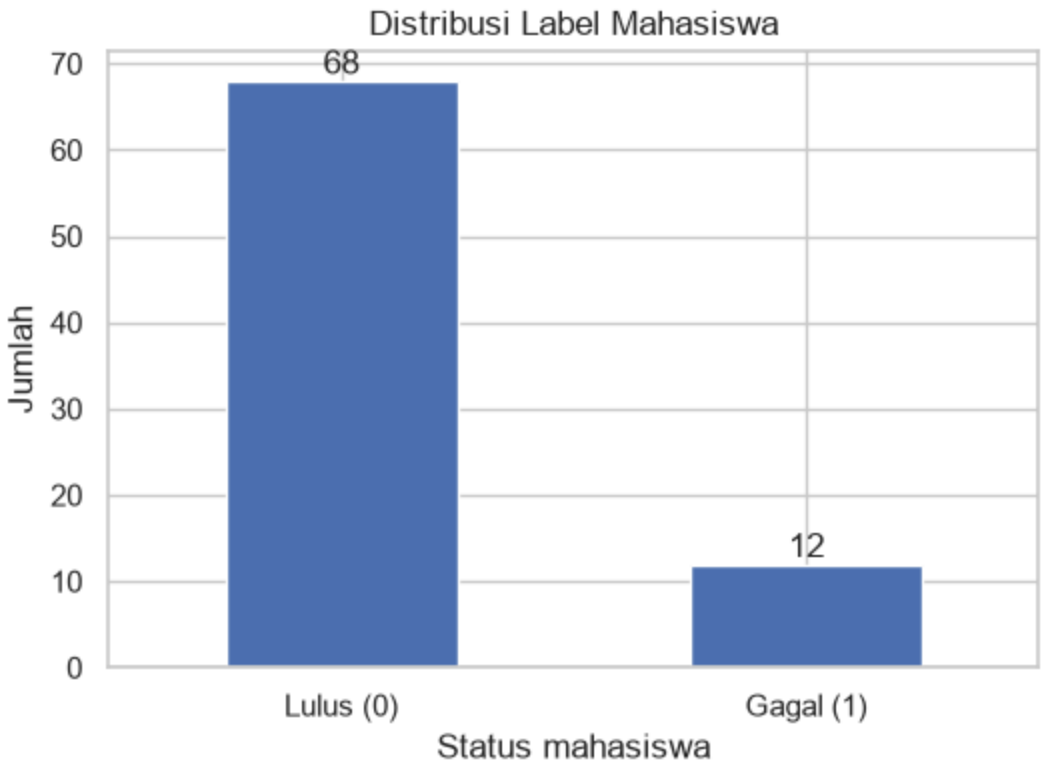
print("Dimensi setelah pembersihan awal:", df.shape)
print("Jumlah duplikat:", df.duplicated().sum())
print("\\nDistribusi label:")
display(df[target_col].value_counts().rename(index={0: "Lulus (0)", 1: "Gagal (1)"}
```

```
label_pct = df[target_col].value_counts(normalize=True).sort_index() * 100
plt.figure(figsize=(6, 4))
counts = df[target_col].map({0: "Lulus (0)", 1: "Gagal (1)"}).value_counts().reinde
ax = counts.plot(kind="bar", rot=0)
ax.set_title("Distribusi Label Mahasiswa")
ax.set_xlabel("Status mahasiswa")
ax.set_ylabel("Jumlah")
for p in ax.patches:
    ax.annotate(f"{int(p.get_height())}", (p.get_x() + p.get_width()/2, p.get_heigh
                ha="center", va="bottom")
plt.show()

print("Persentase kelas:")
display(label_pct.rename(index={0: "Lulus (0)", 1: "Gagal (1)"}).to_frame("Persenta
print("Interpretasi: kelas tidak seimbang karena jumlah mahasiswa lulus jauh lebih
```

```
Dimensi setelah pembersihan awal: (80, 20)
Jumlah duplikat: 0
\nDistribusi label:
```

label (fail=1, pass=0)	Jumlah
Lulus (0)	68
Gagal (1)	12




```
Persentase kelas:
```

	Persentase (%)
label (fail=1, pass=0)	
Lulus (0)	85.0
Gagal (1)	15.0

Interpretasi: kelas tidak seimbang karena jumlah mahasiswa lulus jauh lebih banyak daripada mahasiswa gagal.

EDA

In [13]:
```
#Melakukan pengecekan ulang dan menghapus data yang duplikat
```

In [14]:
```
print("Dimensi data:", df.shape)
print("Jumlah baris duplikat:", df.duplicated().sum())
print("Missing value per kolom:")
display(df.isna().sum().to_frame("Missing"))
```

```
Dimensi data: (80, 20)
Jumlah baris duplikat: 0
Missing value per kolom:
```

	Missing
Gender	0
Age	0
POLAR4 Quintile	0
POLAR3 Quintile	0
Adult HE 2001 Quintile	0
Adult HE 2011 Quintile	0
TUNDRA MSOA Quintile	0
TUNDRA LSOA Quintile	0
Gaps GCSE Quintile	0
Gaps GCSE Ethnicity Quintile	0
Uni Connect target ward	0
attending from home?	0
distance to university (km)	0
Count of Module Area Logins	0
Total Hours in Module Area	0
% of Average Hours in Module Area	0
# of presence	0
# of Absence	2
Percent Attended	0
label (fail=1, pass=0)	0

In [15]:
```
#menghapus data yang duplikat
```

In [16]:
```
df = df.drop_duplicates().copy()
```

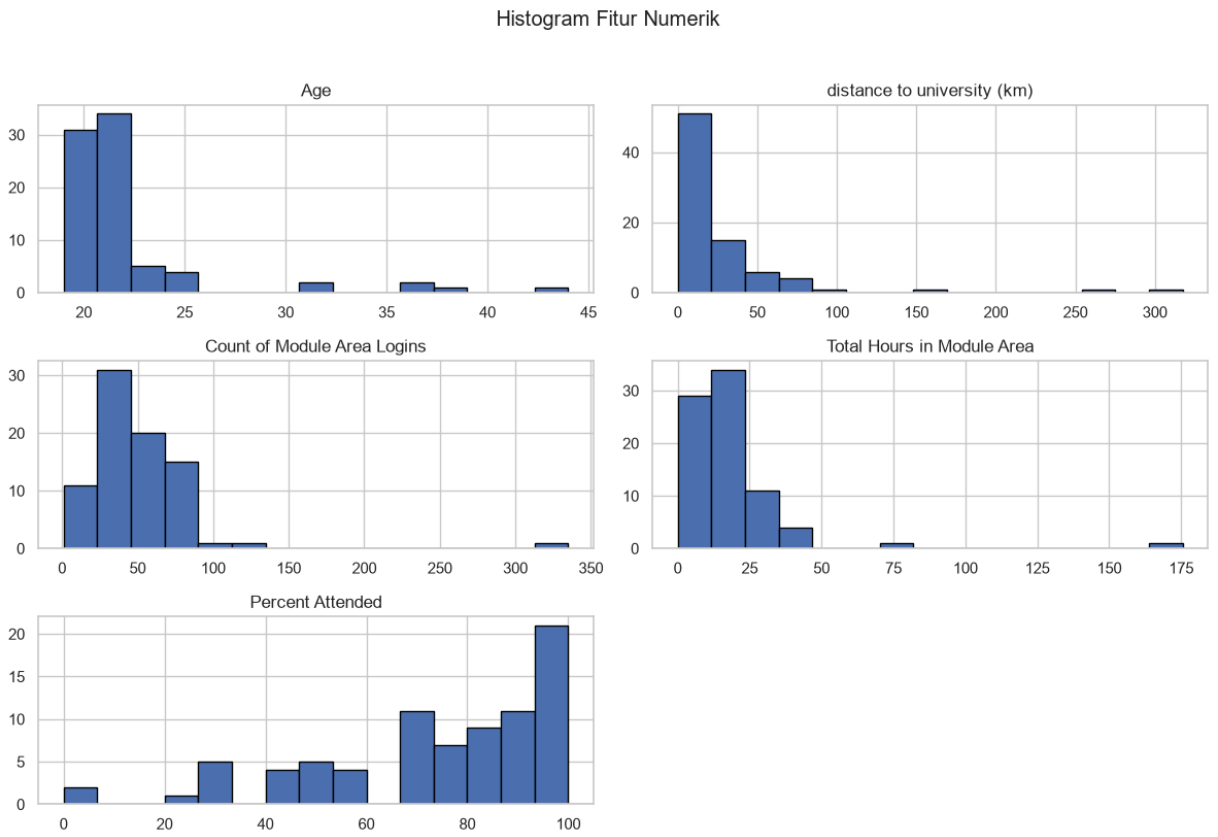
In [17]:
```
#Menyaring kolom kolom numerik dan memisahkannya dengan kolom target
```

In [18]:
```
numeric_cols = df.select_dtypes(include=np.number).columns.tolist()
feature_cols_all = [c for c in numeric_cols if c != target_col]
```

In [19]:
```
#Menampilkan / membuat fitur  histogram
```

In [20]:
```
hist_cols = [
    "Age", "distance to university (km)", "Count of Module Area Logins",
    "Total Hours in Module Area", "Percent Attended"
]
hist_cols = [c for c in hist_cols if c in df.columns]
```

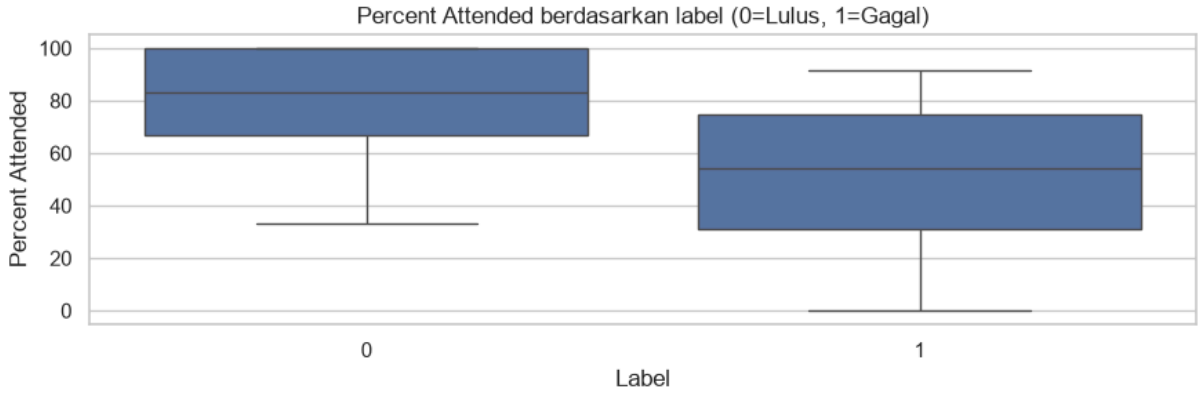
```
df[hist_cols].hist(figsize=(12, 8), bins=15, edgecolor="black")
plt.suptitle("Histogram Fitur Numerik", y=1.02)
plt.tight_layout()
plt.show()
```

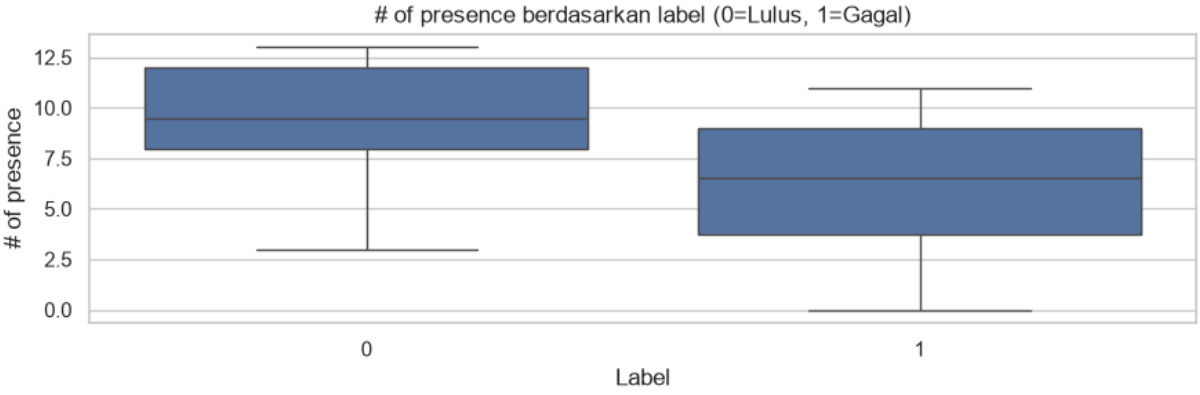


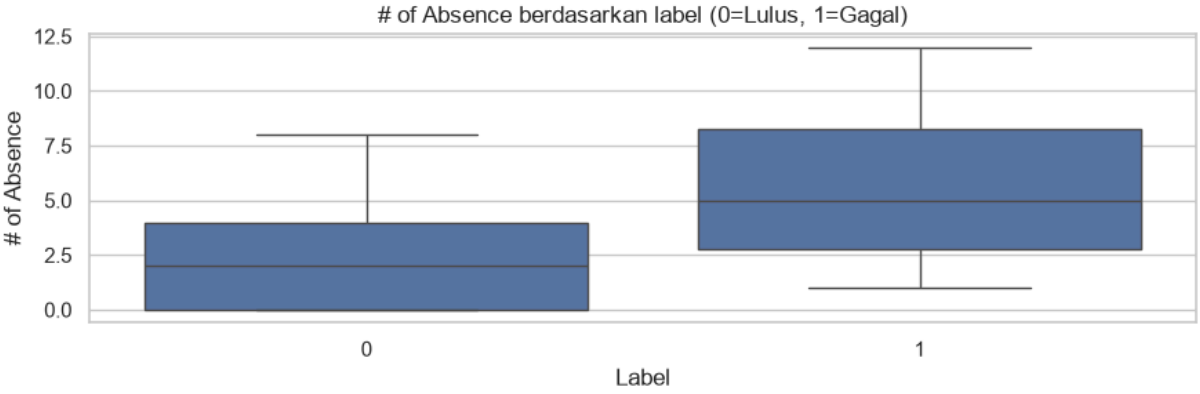
In [21]:
```
#Menghitung nilai korelasi antar fitur
```

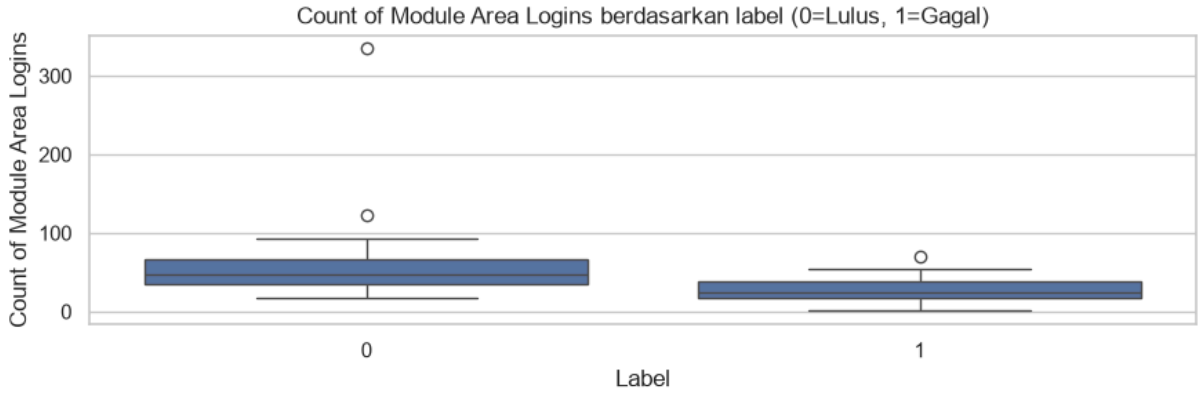
In [22]:
```
box_cols = [
    "Percent Attended", "# of presence", "# of Absence",
    "Count of Module Area Logins", "Total Hours in Module Area"
]
box_cols = [c for c in box_cols if c in df.columns]

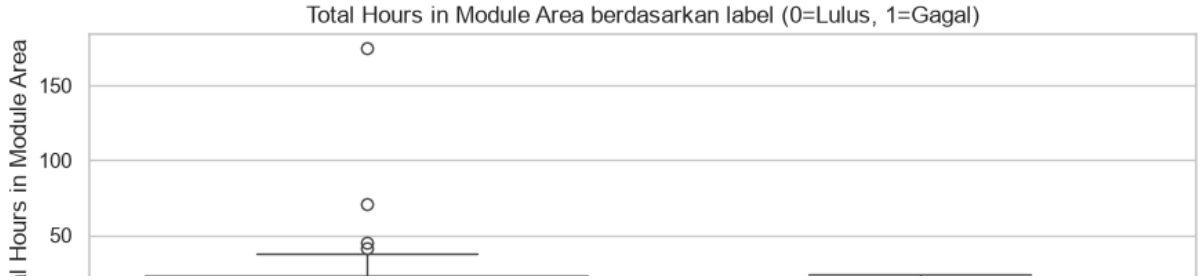
fig, axes = plt.subplots(len(box_cols), 1, figsize=(9, 3 * len(box_cols)))
if len(box_cols) == 1:
    axes = [axes]
for ax, col in zip(axes, box_cols):
    sns.boxplot(data=df, x=target_col, y=col, ax=ax)
    ax.set_title(f"{col} berdasarkan label (0=Lulus, 1=Gagal)")
    ax.set_xlabel("Label")
plt.tight_layout()
plt.show()
```

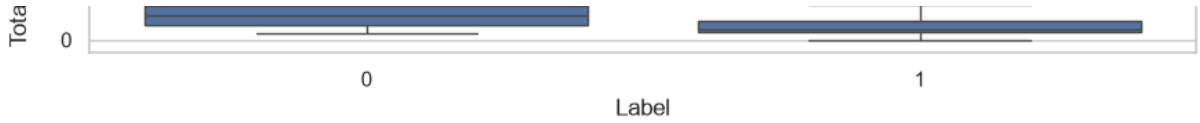

Percent Attended berdasarkan label (0=Lulus, 1=Gagal)
Percent Attended
100
80
60
40
20
0
0
1
Label


of presence berdasarkan label (0=Lulus, 1=Gagal)
of presence
12.5
10.0
7.5
5.0
2.5
0.0
0
1
Label


of Absence berdasarkan label (0=Lulus, 1=Gagal)
of Absence
12.5
10.0
7.5
5.0
2.5
0.0
0
1
Label


Count of Module Area Logins berdasarkan label (0=Lulus, 1=Gagal)
Count of Module Area Logins
300
200
100
0
0
1
Label


Total Hours in Module Area berdasarkan label (0=Lulus, 1=Gagal)
150
100
50

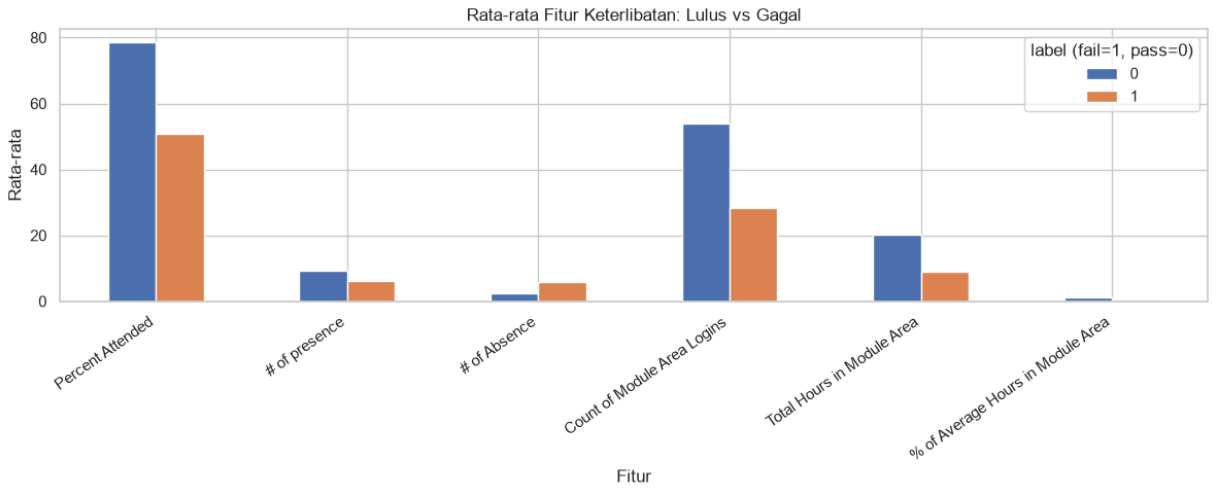


In [23]:
```
#Membuat visualisasi heatmap korelasi
```

In [24]:
```
engagement_cols = [
    "Percent Attended", "# of presence", "# of Absence",
    "Count of Module Area Logins", "Total Hours in Module Area",
    "% of Average Hours in Module Area"
]
engagement_cols = [c for c in engagement_cols if c in df.columns]
mean_by_label = df.groupby(target_col)[engagement_cols].mean().T
display(mean_by_label.rename(columns={0: "Lulus (0)", 1: "Gagal (1)"}))

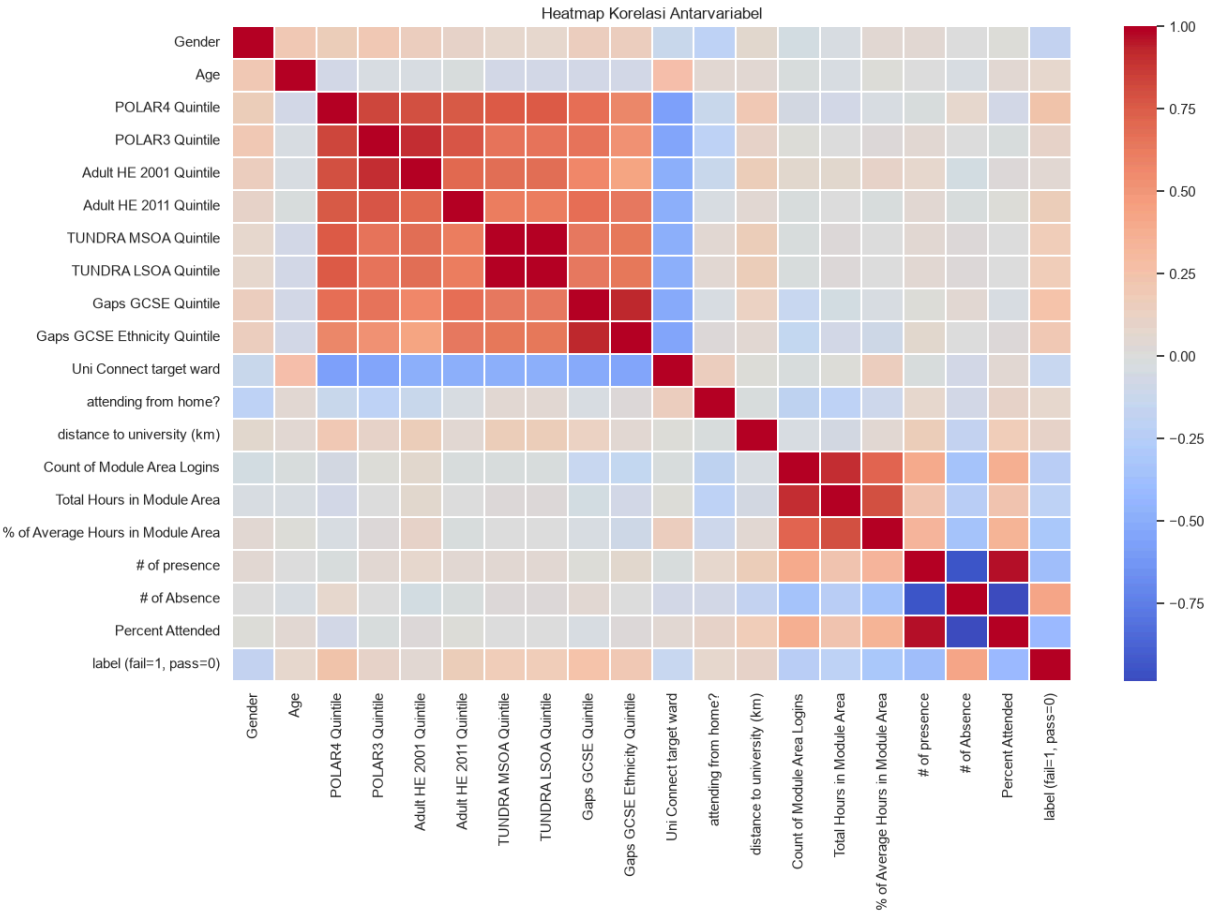
mean_by_label.plot(kind="bar", figsize=(12, 5))
plt.title("Rata-rata Fitur Keterlibatan: Lulus vs Gagal")
plt.ylabel("Rata-rata")
plt.xlabel("Fitur")
plt.xticks(rotation=35, ha="right")
plt.tight_layout()
plt.show()
```

label (fail=1, pass=0)	Lulus (0)	Gagal (1)
Percent Attended	78.553971	50.694444
# of presence	9.161765	6.083333
# of Absence	2.424242	5.750000
Count of Module Area Logins	54.014706	28.416667
Total Hours in Module Area	20.320588	9.023333
% of Average Hours in Module Area	1.186948	0.432312



In [25]:
```
#Membuat plot kotak(boxplot) untuk mencari nilai numerik tertentu
```

In [26]:
```
plt.figure(figsize=(14, 10))
corr = df.select_dtypes(include=np.number).corr()
sns.heatmap(corr, cmap="coolwarm", center=0, annot=False, linewidths=0.3)
plt.title("Heatmap Korelasi Antarvariabel")
plt.tight_layout()
plt.show()
```



In [27]:
```
#Pemrosesan data dan melakukan penanganan lebih lanjut dari outlier
```

In [28]:
```
continuous_cols = [
    "Age", "distance to university (km)", "Count of Module Area Logins",
    "Total Hours in Module Area", "% of Average Hours in Module Area",
    "# of presence", "# of Absence", "Percent Attended"
]
continuous_cols = [c for c in continuous_cols if c in df.columns]

outlier_summary = []
for col in continuous_cols:
    series = df[col].dropna()
    q1, q3 = series.quantile([0.25, 0.75])
    iqr = q3 - q1
    lower = q1 - 1.5 * iqr
    upper = q3 + 1.5 * iqr
    count = ((series < lower) | (series > upper)).sum()
    outlier_summary.append([col, lower, upper, int(count)])
```

```
outlier_df = pd.DataFrame(outlier_summary, columns=["Fitur", "Batas bawah IQR", "Ba
display(outlier_df)
```

	Fitur	Batas bawah IQR	Batas atas IQR	Jumlah outlier
0	Age	17.000000	25.000000	6
1	distance to university (km)	-40.529151	76.240647	7
2	Count of Module Area Logins	-16.125000	108.875000	2
3	Total Hours in Module Area	-11.981250	42.768750	3
4	% of Average Hours in Module Area	-0.474730	2.343004	4
5	# of presence	1.000000	17.000000	2
6	# of Absence	-6.500000	11.500000	2
7	Percent Attended	-4.166667	162.500000	0

In [29]:
```
#menampilkan jumlah nilai pada fitur kuintil
```

In [30]:
```
quintile_cols = [c for c in df.columns if "Quintile" in c]
zero_quintiles = {c: int((df[c] == 0).sum()) for c in quintile_cols}
print("Jumlah nilai 0 pada fitur kuintil:")
display(pd.Series(zero_quintiles, name="Jumlah nilai 0").to_frame())

print("Keputusan: nilai outlier numerik dipertahankan karena belum terbukti salah."
print("Nilai 0 pada kolom kuintil diperlakukan sebagai nilai tidak tersedia (NaN),
```

```
Jumlah nilai 0 pada fitur kuintil:
```

	Jumlah nilai 0
POLAR4 Quintile	0
POLAR3 Quintile	0
Adult HE 2001 Quintile	0
Adult HE 2011 Quintile	2
TUNDRA MSOA Quintile	2
TUNDRA LSOA Quintile	2
Gaps GCSE Quintile	4
Gaps GCSE Ethnicity Quintile	4

```
Keputusan: nilai outlier numerik dipertahankan karena belum terbukti salah.
Nilai 0 pada kolom kuintil diperlakukan sebagai nilai tidak tersedia (NaN), karena d
okumen tugas menyatakan maknanya tidak dijelaskan dan mungkin menandakan data tidak
tersedia.
```

PERSIAPAN DATA

In [31]:
```
#Mengkopi dataframe ke dalam model_df
```

In [32]:
```
model_df = df.copy()
```

In [33]:
```
#Mengubah nilai 0 pada kolom Quintile menjadi missing value
```

In [34]:
```
quintile_cols = [c for c in model_df.columns if "Quintile" in c]
for col in quintile_cols:
    model_df.loc[model_df[col] == 0, col] = np.nan
```

In [35]:
```
#Menghapus kolom yang tidak diperlukan
```

In [36]:
```
redundant_cols = ["TUNDRA LSOA Quintile", "Percent Attended"]
drop_cols = [c for c in redundant_cols if c in model_df.columns]
model_df = model_df.drop(columns=drop_cols)

X = model_df.drop(columns=[target_col])
y = model_df[target_col].astype(int)
```

In [37]:
```
#Menampilkan informasi singkat mengenai ukuran dan distribusi data tersebut
```

In [38]:
```
X = X.apply(pd.to_numeric, errors="coerce")

print("Fitur yang digunakan:", X.columns.tolist())
print("Jumlah fitur:", X.shape[1])
print("Ukuran X:", X.shape, "| Ukuran y:", y.shape)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=RANDOM_STATE, stratify=y
)

print("Data latih:", X_train.shape, "| Data uji:", X_test.shape)
print("\\nDistribusi label data latih:")
display(y_train.value_counts().rename(index={0: "Lulus", 1: "Gagal"}).to_frame("Jum
print("\\nDistribusi label data uji:")
display(y_test.value_counts().rename(index={0: "Lulus", 1: "Gagal"}).to_frame("Juml
```

```
Fitur yang digunakan: ['Gender', 'Age', 'POLAR4 Quintile', 'POLAR3 Quintile', 'Adult
HE 2001 Quintile', 'Adult HE 2011 Quintile', 'TUNDRA MSOA Quintile', 'Gaps GCSE Quin
tile', 'Gaps GCSE Ethnicity Quintile', 'Uni Connect target ward', 'attending from ho
me?', 'distance to university (km)', 'Count of Module Area Logins', 'Total Hours in
Module Area', '% of Average Hours in Module Area', '# of presence', '# of Absence']
Jumlah fitur: 17
Ukuran X: (80, 17) | Ukuran y: (80,)
Data latih: (64, 17) | Data uji: (16, 17)
\nDistribusi label data latih:
```

label (fail=1, pass=0)	Jumlah
Lulus	54
Gagal	10

\nDistribusi label data uji:

label (fail=1, pass=0)	Jumlah
Lulus	14
Gagal	2

MODELING

In [39]:
```
#Mempersiapkan metode validasi dan mengevaluasi masing masing model
```

In [40]:
```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)

models = {
    "Logistic Regression": Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler()),
        ("model", LogisticRegression(class_weight="balanced", max_iter=2000, random
    ]),
    "Decision Tree": Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("model", DecisionTreeClassifier(class_weight="balanced", random_state=RAND
    ]),
    "Random Forest": Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("model", RandomForestClassifier(
            n_estimators=300, class_weight="balanced",
            random_state=RANDOM_STATE, max_depth=4, min_samples_leaf=2
        ))
    ])
}

cv_rows = []
for name, pipeline in models.items():
    scores = cross_validate(
        pipeline, X_train, y_train, cv=cv,
        scoring={"f1": "f1", "recall": "recall", "precision": "precision", "accurac
        error_score="raise"
    )
    cv_rows.append({
        "Model": name,
        "CV F1 gagal (rata-rata)": scores["test_f1"].mean(),
        "CV Recall gagal (rata-rata)": scores["test_recall"].mean(),
        "CV Precision gagal (rata-rata)": scores["test_precision"].mean(),
        "CV Accuracy (rata-rata)": scores["test_accuracy"].mean()
    })

cv_results = pd.DataFrame(cv_rows).sort_values("CV F1 gagal (rata-rata)", ascending
display(cv_results)
```

	Model	CV F1 gagal (rata-rata)	CV Recall gagal (rata-rata)	CV Precision gagal (rata-rata)	CV Accuracy (rata-rata)
0	Logistic Regression	0.493333	0.6	0.483333	0.798718
1	Decision Tree	0.413333	0.5	0.416667	0.798718
2	Random Forest	0.360000	0.4	0.383333	0.767949

In [41]:
```
#Mencari kombinasi parameter terbaik
```

In [42]:
```
rf_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("model", RandomForestClassifier(class_weight="balanced", random_state=RANDOM_S
])

param_grid = {
    "model__n_estimators": [200, 300],
    "model__max_depth": [None, 3, 5],
    "model__min_samples_leaf": [1, 2, 3]
}

grid_search = GridSearchCV(
    rf_pipeline, param_grid=param_grid, scoring="f1",
    cv=cv, n_jobs=-1, refit=True
)
grid_search.fit(X_train, y_train)

print("Parameter terbaik:", grid_search.best_params_)
print("F1 CV terbaik untuk kelas gagal:", round(grid_search.best_score_, 4))
```

```
Parameter terbaik: {'model__max_depth': None, 'model__min_samples_leaf': 1, 'model__
n_estimators': 200}
F1 CV terbaik untuk kelas gagal: 0.4133
```

In [43]:
```
#melatih seluruh model awal
```

In [44]:
```
fitted_models = {}
for name, pipeline in models.items():
    pipeline.fit(X_train, y_train)
    fitted_models[name] = pipeline
fitted_models["Random Forest Tuned"] = grid_search.best_estimator_

print("Semua model berhasil dilatih.")
```

```
Semua model berhasil dilatih.
```

EVALUASI MODEL

In [45]:
```
#Mengevaluasi seluruh model yang telah dilatih
```

In [46]:
```
evaluation_rows = []
for name, model in fitted_models.items():
    y_pred = model.predict(X_test)
    evaluation_rows.append({
        "Model": name,
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision gagal": precision_score(y_test, y_pred, zero_division=0),
        "Recall gagal": recall_score(y_test, y_pred, zero_division=0),
        "F1 gagal": f1_score(y_test, y_pred, zero_division=0)
    })

evaluation = pd.DataFrame(evaluation_rows).sort_values("F1 gagal", ascending=False)
display(evaluation)

best_model_name = evaluation.loc[0, "Model"]
best_model = fitted_models[best_model_name]
print("Model terbaik berdasarkan F1 kelas gagal pada data uji:", best_model_name)
```

	Model	Accuracy	Precision gagal	Recall gagal	F1 gagal
0	Random Forest Tuned	0.9375	1.00	0.5	0.666667
1	Random Forest	0.9375	1.00	0.5	0.666667
2	Logistic Regression	0.7500	0.25	0.5	0.333333
3	Decision Tree	0.8125	0.00	0.0	0.000000

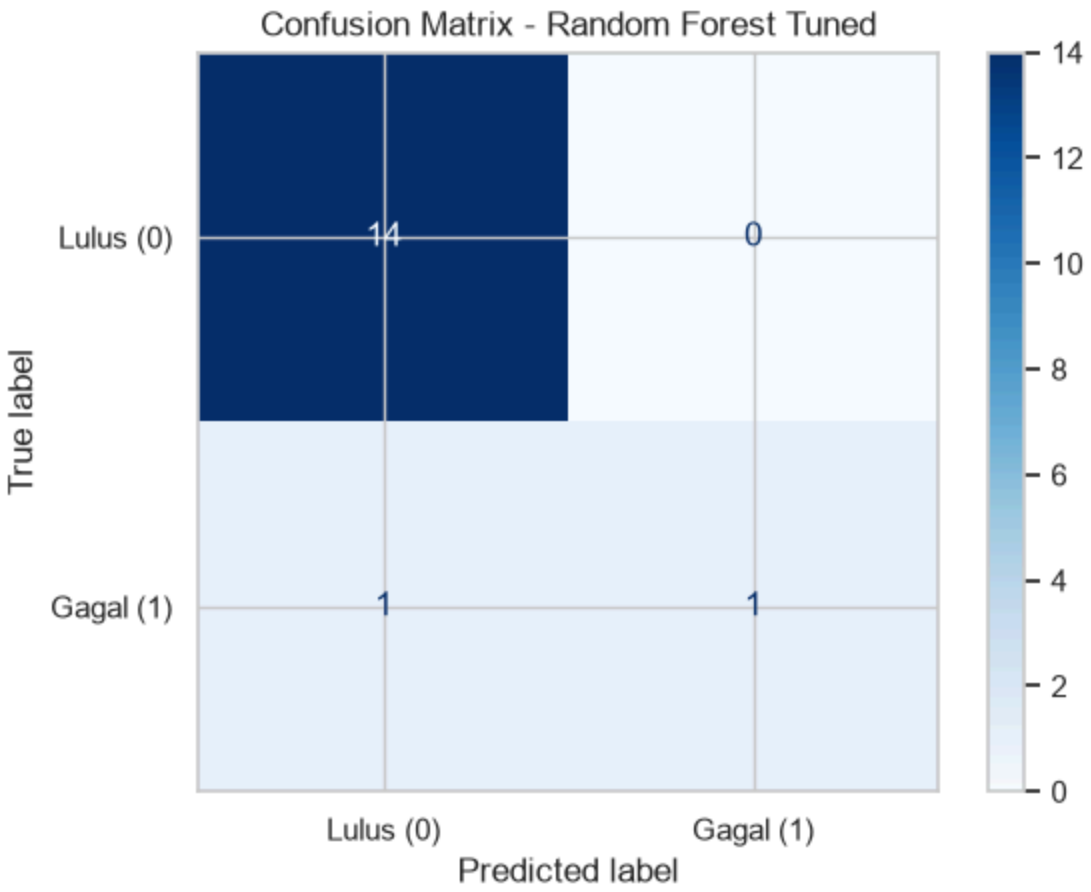
```
Model terbaik berdasarkan F1 kelas gagal pada data uji: Random Forest Tuned
```

In [47]:
```
#Mengevaluasi performa model terbaik secara mendalam dan mencetak laporan klasifika
```

In [48]:
```
best_pred = best_model.predict(X_test)
cm = confusion_matrix(y_test, best_pred, labels=[0, 1])

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Lulus (0)", "Gagal (1)"]
)
disp.plot(cmap="Blues", values_format="d")
plt.title(f"Confusion Matrix - {best_model_name}")
plt.show()

print("Classification Report:")
print(classification_report(
    y_test, best_pred, labels=[0, 1],
    target_names=["Lulus (0)", "Gagal (1)"],
    zero_division=0
))
```




```
Classification Report:
              precision    recall  f1-score   support

   Lulus (0)       0.93      1.00      0.97        14
   Gagal (1)       1.00      0.50      0.67         2

    accuracy                           0.94        16
   macro avg       0.97      0.75      0.82        16
weighted avg       0.94      0.94      0.93        16
```

In [49]:
```
#Mengekstrak dan memvisualisasikan tingkat kepentingan fitur
```

In [50]:
```
final_estimator = best_model.named_steps["model"]

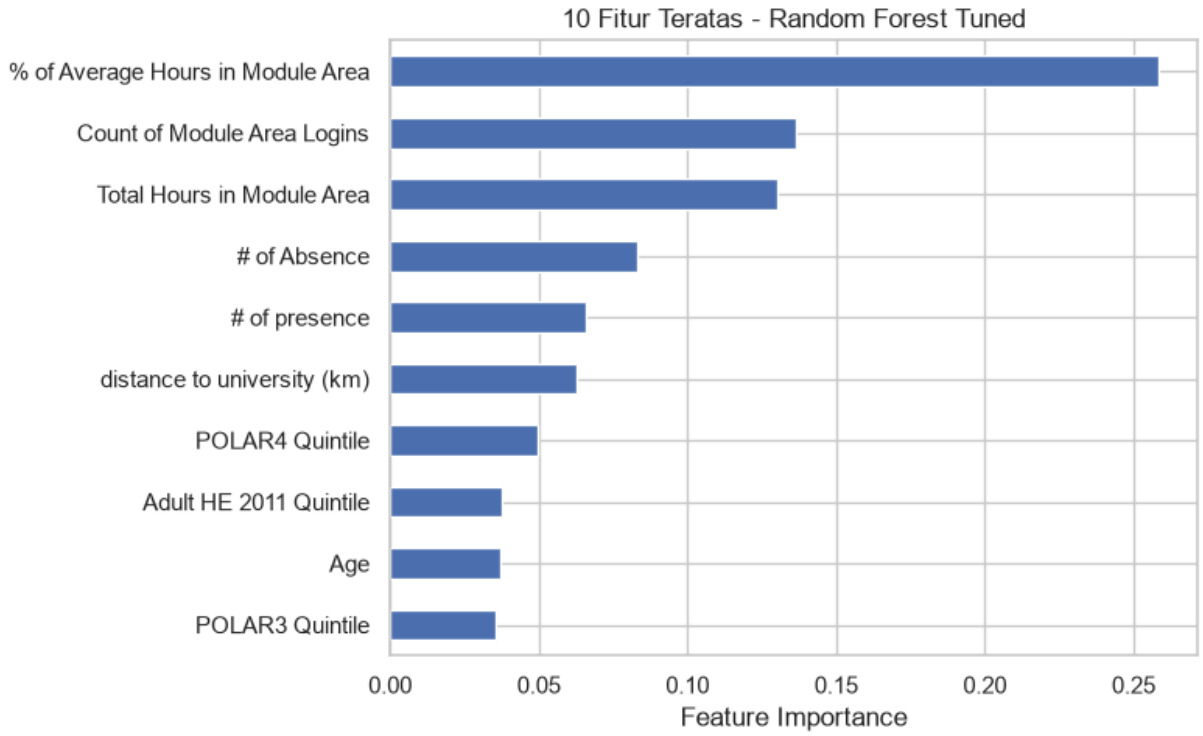
if hasattr(final_estimator, "feature_importances_"):
    importance = pd.Series(final_estimator.feature_importances_, index=X.columns).s
    importance_name = "Feature Importance"
elif hasattr(final_estimator, "coef_"):
    importance = pd.Series(np.abs(final_estimator.coef_[0]), index=X.columns).sort_
    importance_name = "Nilai absolut koefisien Logistic Regression"
else:
    importance = pd.Series(dtype=float)
    importance_name = "Tidak tersedia"

if not importance.empty:
    print(importance_name)
    display(importance.to_frame("Nilai").head(10))
    importance.head(10).sort_values().plot(kind="barh", figsize=(8, 5))
```

```
    plt.title(f"10 Fitur Teratas - {best_model_name}")
    plt.xlabel(importance_name)
    plt.tight_layout()
    plt.show()
else:
    print("Model ini tidak menyediakan feature importance atau koefisien.")
```

```
Feature Importance
```

	Nilai
% of Average Hours in Module Area	0.258540
Count of Module Area Logins	0.136323
Total Hours in Module Area	0.130199
# of Absence	0.082942
# of presence	0.065965
distance to university (km)	0.062835
POLAR4 Quintile	0.049676
Adult HE 2011 Quintile	0.037699
Age	0.037104
POLAR3 Quintile	0.035377



Tes Prediksi Dengan Data Random

In [51]:
```
#Membuat generator angka acak
```

In [52]:
```
rng = np.random.default_rng(RANDOM_STATE)
random_students = {}
```

In [53]:
```
#membuat daata sampel / data buatan untuk 5 mahasiswa baru tiruan
```

In [54]:
```
discrete_cols = [
    "Gender", "POLAR4 Quintile", "POLAR3 Quintile",
    "Adult HE 2001 Quintile", "Adult HE 2011 Quintile",
    "TUNDRA MSOA Quintile", "Gaps GCSE Quintile",
    "Gaps GCSE Ethnicity Quintile", "Uni Connect target ward",
    "attending from home?"
]
discrete_cols = [c for c in discrete_cols if c in X.columns]

for col in X.columns:
    values = X[col].dropna()
    if values.empty:
        random_students[col] = np.zeros(5)
    elif col in discrete_cols:
        choices = values.unique()
        random_students[col] = rng.choice(choices, size=5, replace=True)
    else:
        low, high = values.min(), values.max()
        random_students[col] = rng.uniform(low, high, size=5)

new_students = pd.DataFrame(random_students, columns=X.columns)
display(new_students.round(2))
```

	Gender	Age	POLAR4 Quintile	POLAR3 Quintile	Adult HE 2001 Quintile	Adult HE 2011 Quintile	TUNDRA MSOA Quintile	Gaps GCSE Quintile	Gaps GCSE Ethnicity Quintile	U Conne targ wa
0	0.0	36.43	2.0	4.0	2.0	4.0	4.0	5.0	2.0	1
1	1.0	21.35	3.0	2.0	3.0	3.0	5.0	1.0	1.0	0
2	1.0	43.39	5.0	5.0	5.0	5.0	5.0	1.0	2.0	0
3	0.0	38.03	2.0	1.0	3.0	4.0	3.0	4.0	3.0	0
4	0.0	38.65	3.0	3.0	3.0	2.0	1.0	5.0	3.0	0

In [55]:
```
#Melakukan prediksi kelulusan dan menghitung peluang dari 5 mahasiswa tiruan
```

In [56]:
```
new_predictions = best_model.predict(new_students)
new_probabilities = best_model.predict_proba(new_students)
```

In [57]:
```
#Menggabungkan, merapikan, dan mengubah hasil prediksi mentah menjadi tabel laporan
```

In [58]:
```
class_order = list(best_model.classes_)
fail_index = class_order.index(1)
fail_probability = new_probabilities[:, fail_index]

prediction_result = new_students.copy()
prediction_result["Prediksi label"] = new_predictions
prediction_result["Prediksi status"] = np.where(new_predictions == 1, "GAGAL", "LUL
prediction_result["Probabilitas gagal (%)"] = (fail_probability * 100).round(2)
prediction_result["Probabilitas lulus (%)"] = ((1 - fail_probability) * 100).round(

display(prediction_result.round(2))
```

	Gender	Age	POLAR4 Quintile	POLAR3 Quintile	Adult HE 2001 Quintile	Adult HE 2011 Quintile	TUNDRA MSOA Quintile	Gaps GCSE Quintile	Gaps GCSE Ethnicity Quintile	U Conne targ wa
0	0.0	36.43	2.0	4.0	2.0	4.0	4.0	5.0	2.0	1
1	1.0	21.35	3.0	2.0	3.0	3.0	5.0	1.0	1.0	(
2	1.0	43.39	5.0	5.0	5.0	5.0	5.0	1.0	2.0	(
3	0.0	38.03	2.0	1.0	3.0	4.0	3.0	4.0	3.0	(
4	0.0	38.65	3.0	3.0	3.0	2.0	1.0	5.0	3.0	(

In [ ]:
```
```